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sermon on the mountain



As part of his consulting stint at Bose, Tufte reviewed a new language-less setup guide for the Wave Radio, below, and offered more than 100 suggestions, which were incorporated almost immediately.

How Edward Tufte led Bose out of the land of chartjunk.

by James Surowiecki

"A t one time or another, I've consulted for just about everybody," Edward Tufte says, mentioning names like IBM, Hewlett-Packard, and Sun Microsystems. "But I don't really do much of it anymore. It's too big an effort for too small a result." Since the early 1980s, when Tufte, a professor of political science, statistics, and computer science at Yale, started writing about information design, corporate America has been relentlessly seeking out his advice. But it wasn't until he arrived at the Bose Corporation in 1995 that he found a company that was actually ready, as one of his champions there puts it, to be "Tufte-ized."

"Trying to change IBM is like trying to change Sweden," Tufte says. "It's difficult when big systems are designed because there are a lot of thumbprints left on them, and things just get negotiated away. It's very hard to have a real impact." Bose, on the other hand, is a small, privately held company, founded and still run by Amar Bose, an electrical engineering professor at MIT whose obsession with great sound is nearly matched by his fascination with design. All of Bose's profits are reinvested in research, and no one has to worry about keeping the stockholders happy by skimping on product quality or design. "It feels more like an engineering department at a big university than a typical corporate environment," Tufte says. Bose remade everything from its instruction manuals and Web site to its corporate-finance presentations and product-design process according to Tufte's philosophy, which holds that the principles of good design are universal, and can be applied to every medium.

Tufte, a lanky man who, at 56, speaks with an appealing mix of diffidence and self-assurance, might be said to have invented information design as a formal discipline. His self-published trilogy of books on the subject, The Visual Display of Quantitative Information (1983), Envisioning Information (1989), and Visual Explanations (1997), eloquently articulates the fundamentals of successful information design while critiquing the ways in which designers of all stripes deploy maps, charts, and pictures. Tufte believes that the best visual presentation of statistical data communicates content efficiently and coherently without resorting to embellishment or overkill. He emphasizes that good designs reveal data "at several levels of detail" — ranging from broad overview to fine structure — and that they present comparisons with other data sets. But he also warns that graphics can be misleading if they attempt to compare incomparable data sets or make the visual differences between numbers appear larger than the actual differences, or if they obscure changes over time.

Tufte's premise is that the clearer and more elegant the presentation of information, the greater the chance that it will be used well. As he puts it, "Superior methods [for displaying and analyzing data] are more likely to produce truthful, credible, and precise findings." His hostility to what he calls "chartjunk" — "ink that does not tell the viewer anything new" — is not simply an aesthetic aversion, but stems from his conviction that whether the phenomenon occurs in the form of gratuitous effects or extraneous color, it prevents people from gleaning the very information that the chart purports to offer. Chartjunk, says Tufte, generates "no information, no sense of discovery, no wonder, no substance." The simplicity of Tufte's questions — "Is the display revealing the truth?" "Are appropriate comparisons and contexts shown?" — points to his belief that an information designer's sole allegiance should be to "the integrity of the content displayed." While this might appear to limit designers' styles to the routine and predictable, the reality ends up being very different. Tufte's books are lavishly illustrated with "successful" designs that are also quite beautiful and varied.

In the last decade, these books have become what Fortune calls "bibles of design for thousands of engineers, computer designers, scientists, and financial analysts." Nearly 500,000 copies of the three volumes are in print, and Tufte is one of Amazon.com's Top 100 authors. Hundreds of people regularly pack his one-day seminars, paying $300 to hear him. When you consider the amount of time and money that corporations spend on the presentation of information, whether employee to employee, business to business, or producer to consumer, it's not surprising that corporations would be eager to hire him. But as his comments about IBM suggest, wanting Tufte to consult for you is not the same as allowing his principles to inform your business from the ground up.

At Bose, though, Tufte's ideas — on both the larger relationship between design and content and the more specific need for clear, uncluttered presentations — resonated immediately. The company had always placed a premium on a seamless design process. Instead of treating product design, sound engineering, and manufacturing as self-contained divisions, Bose wanted to fully integrate the creation of its products from the beginning. You don't invent objects without thinking about how to mass-produce them and how to ensure they'll be user-friendly. "I felt that there might be a lot that he could offer in a consultative role here at Bose, because his work ranges across so many fields," says Ken Jacob, director of the company's professional sound division, speaking from his office on the Mountain, the glacial formation in Framingham, Massachusetts, where Bose is located. "But he really resisted at first, because I think his efforts in consulting had left him with a bitter taste in his mouth. He had worked very hard with a few companies, for what he felt was very little result."

Jacob had read Tufte's books and attended one of his seminars in 1994. "After reading the books cover to cover — I mean every square centimeter — I realized that Tufte was on to something that was just huge," Jacob says exuberantly. "And I thought he was one of the few guys on the planet who would make it worth my while to sit at his feet and listen."

By the time Jacob convinced Tufte to come to Bose as a consultant, nearly three years ago, the company culture was primed for his guidance. Thanks in no small part to Jacob's proselytizing, hundreds of Bose employees had already attended Tufte's seminar. While his arrival was not quite equivalent to Moses coming down from Mount Sinai, it was close enough. During his visits, Tufte played the role of a teacher, critiquing presentations made by employees in nearly every area of the company.

Some of the professor's advice was textbook Tufte. He lambasted the company's Web site for devoting far too many pixels to navigation as opposed to content, emphasizing the need to keep a high data-to-ink (well, figurative ink) ratio on each page. Along the same lines, he stressed the value of presenting information spatially — that is, putting as much information as possible on one screen — rather than temporally, thus averting the "one damn screen after another" curse that afflicts so much Web design. Today, while the opening page of Bose's site

(www.bose.com) is still heavy on color and hard-to-read images, the rest of the site is clean and content-rich. Tufte also reviewed a new language-less setup guide for the Wave Radio and rattled off more than 100 suggestions for improvement, which were incorporated almost immediately. The guide went on to win the 1998 Award for Excellence of Art Design at the International Technical Art Competition.

But during his tenure at Bose, Tufte also emphasized that good information design depends on considerations beyond the realm of design. When he was handed a chart that showed the progress of different projects compared with the initial plans for those projects — faithfully designed according to Tuftean principles — he asked simply whether the cost of the measurements needed to make the design work was worth the benefits derived from it. The answer, it turned out, was "No." Tufte's question was the perfect expression of his belief that a design can never be better than the information that goes into it. Similarly, upon reviewing a user interface for a new product (one that is now on the verge of being released, and therefore is a closely guarded secret), Tufte asked how the designers could predict that the interface technology would not be obsolete by the time the product was brought to market. Again, the question illuminated his emphasis on appropriate comparisons — thinking not about how much different technologies cost now, but rather how much they will ultimately cost — and his focus on changes in data over time. The research team eventually concluded that the interface would not be obsolete, but tackling that question helped Bose reorient its research to anticipate future technologies.

Although his consulting gig is over, Tufte's influence at Bose is, if anything, on the rise. "He has made a real impact on this place. A company with hundreds of people thinking differently about design and looking to make sure that such thinking results in better products is going to have an edge," Jacob says. "There are innumerable influences on a corporation the size of Bose, but Tufte is certainly one of the influences that results in a competitive advantage."

Today, when design is being touted by many as a magic bullet — gonzo management guru Tom Peters insists in his book The Circle of Innovation (1997) that "design is the Big Enchilada . . . in fact the WHOLE Enchilada" — it's important to remember that the exaltation of design, like the exaltation of "brand" that often accompanies it, can obscure the competitive benefits companies reap from more mundane advantages like production-line efficiencies, hefty R&D budgets, and well-organized distribution networks. Still, the computer that sits on my desktop is there only because Toshiba had the bright idea of coming out with an all-black PC, and similar design-motivated impulses were behind almost every household purchase I've made in recent years. In that very simple sense, then, design clearly matters.

But there's also a deeper sense in which design matters. Tufte has given Bose a more profound awareness of how to make its many parts one. He has taught employees to operate with more respect for each other — no more 30-page, slide-laden reports — and with more attention to how the people who buy their products use them. As Tufte himself goes to great pains to point out, Bose has always been a corporation that thought about these issues. Perhaps, then, what Tufte helped Bose do was to realize itself more fully, to rediscover that applying the right principles of design can yield far more than a beautiful new stereo.

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